

Origins and Evolution of Language
Week 2 workshop
Tutor notes

Comments for tutors are in italics.

Basics of evolutionary theory

The first tutorial will be hands-on, to give you a chance to check your understanding of the basics of the comparative method and evolution by natural selection.

Phylogenetic trees

Using the [Evolution Lab game](#) (click “play game”), complete at least the three training trees (Red, green and gecko; Familiar faces; Tree of life: Vegetarian edition). The aim here is to understand why closely related species might be expected to share many traits, and how patterns of shared and differing traits between organisms can be informative about evolutionary history of traits and species (e.g. the patterns of relatedness between species, and when certain traits are likely to have evolved).

I would suggest splitting them into several small groups to work on this exercise and the next one, and either go round and talk to the individual groups as they work through, or reconvene as a big group after each task.

They need to use the magnifying glass to examine traits to do this task sensibly.

Things to learn:

- *Species inherit traits from their common ancestor, so if two species share a trait they are likely to share a common ancestor who had that trait. NB They didn't “come from” that common ancestor though - something that people often get confused about is whether humans “come from” monkeys or chimps; no, we share a common ancestor with them, who is likely to have had the traits we share with chimps.*
- *The distribution of traits can therefore help us understand the relationships between species (species that share more traits probably have a more recent common ancestor) **and** the evolution of those traits (specifically, which traits are recent innovations versus ancestral traits; how recent are those traits).*
- *The tutorial is also designed to help you understand that intuitive groupings (e.g. grouping mushrooms and palm trees together in the first practice tree; grouping dog and fish together in the second; grouping banana and lemon together in the 3rd) may be wrong if they don't capture the pattern of shared traits.*
- *People might ask why you can't have reversals or independent evolution of traits – e.g. in the mushroom-gecko-palm tree tree, couldn't “heterotrophic” evolve twice, once in geckos and once independently in mushrooms. Yes it could, but the idea is that in general the evolutionary tree that involves the fewest independent innovations and independent losses etc is most likely to be right.*
- *One weird thing about the way they have set up these trees is you can't put multiple traits on the root node – e.g. in the same mushroom-gecko-palm tree, I wanted to put “heterotrophic” on the root node and then have it replaced on the palm node. I*

think that's just a design choice on the app, you can have loss of traits as well as gain of traits in the real world!

- *Hopefully they will make the connection to linguistics and the reconstruction of protolanguages and language phylogenies through shared traits themselves, but if not you can point this out – in fact linguists were the first ones doing comparative reconstruction! See e.g.*

<https://academic.oup.com/sysbio/article/54/4/513/2842862/> which provides a historical overview.

Evolution by natural selection

Use the interactive simulator linked below (from a suite of simulation tools provided by the PhET team at the University of Colorado Boulder) to run the four experiments below. The aim here is to get a basic understanding of the effects of selection on frequencies of traits (and genes) in a population.

[Use this link to access the simulation](#), then click “Intro” and then “Play” to begin. You can speed things up or restart the simulation by clicking the buttons in the bottom right.

If the students come out of this exercise understanding that, in general, variants which are better adapted to their environment win out, that is enough!

Behind the scenes in this simulation, each bunny has two parents and inherits two versions (alleles) of a gene that affects fur colour, one from each parent. There are two possible alleles: a brown allele and a white allele. If a bunny inherits two white alleles, it has white fur; if it inherits at least one brown allele, it has brown fur (i.e. in technical terms, the brown allele is dominant and the white fur allele is recessive).

You might need to talk them through this. So if B is the brown allele and w is the white allele, there are 4 possible combinations.

BB (B inherited from both parents)

Bw or wB (B inherited from 1 parent, w from the other)

ww (w inherited from both parents)

In the underlying model, I am assuming that bunnies are paired to reproduce, and each parent contributes one of its alleles at random (e.g. a BB individual passes on B to its offspring, a Bw individual might pass on B or w). This is only important in as much as it explains how you can sometimes have w alleles “hiding” in a population of brown bunnies – some brown bunnies might be Bw or wB.

Use this simulation to run these experiments.

- Experiment 1: Starting with the population of 5 brown and 5 white bunnies, predict what will happen to the population over time (say five generations). Run it to find out if you were right.

They might expect that the population hits a natural plateau of population size, but in fact that doesn't happen – (see experiment 2) – the bunnies breed like rabbits and their numbers explode. Secondly, and more subtly, they might expect the numbers of brown and white bunnies to remain roughly equal (it starts at 5-5 after all). But in fact the proportion stabilises on 75% brown, 25% white. This is because the B and w alleles are equally frequent in the initial population (it's 5 BB and 5 ww individuals). Mating is random in the model, so the offspring are equally likely to be BB, Bw, wB and ww, so the gene frequencies don't change generation to generation, B and w stay equally frequent. But of course, because w is recessive, only the 25% of bunnies with the ww genotype actually have white fur. Understanding the maths of this is not important– the key thing is just to understand that the proportion of brown/white bunnies depends a bit on the dominance/recessive relationship between the genes, and you can have w genes hiding in brown bunnies.

- Experiment 2: What do you predict will happen if you select the “Limited food” option? Restart the simulation, click “Limited food”, and run it.

Limiting food means that there's a limit on the maximum population size – so when the bunnies become too numerous (above 300, it looks like) there is a die-off and the population drops then recovers. Not that fur colour doesn't affect this die-off – it seems like the proportion of brown and white bunnies, B and w alleles, isn't affected by the food limit.

- Experiment 3: Repeat Experiment 1, but introduce wolves. Predict what happens to the frequency of brown fur, then run it.

“Repeat Experiment 1” = turn off food limits. When the wolves come in they tend to eat the white bunnies, so the brown bunnies are more likely to survive, reproduce, and pass on their B alleles to the next generation. So brown bunnies take over the population as the w genes carried by the white bunnies are selected against.

- Experiment 4: Repeat Experiment 3, but switch from the sunny to snowy environment (sun vs snowflake icon). Predict what happens to the frequency of brown fur. Does your prediction change? Why? Run it to find out what happens.

This just flips – in the snowy environment the brown bunnies are more likely to be eaten, so the B genes are selected against.

- Experiment 5: Start in the sunny environment, add wolves, and run the simulation for 5 generations. Then flip to the snowy environment and pause the simulation. What do you predict will happen when you restart the simulation? Run it to find out if you were right. NB. this simulation doesn't always play out the same way, so run it a few times and see if you can figure why not!

For the first 5 generations the wolves preferentially eat the white bunnies, removing the w allele from the population by eating bunnies carrying a double copy of it (ww bunnies, who are white). When the environment changes, the previously-well-adapted brown

bunnies are now poorly adapted, and start getting eaten by the wolves. Often, this results in the population being entirely wiped out.

But sometimes, after 5 generations of selection against the w allele in the sunny environment, there are still a few individuals with w alleles floating about in the population (Bw or wB genotypes) – these bunnies are brown, because the w gene is recessive, but if they reproduce with another Bw or wB bunny they can produce a ww offspring. If that happens, their ww offspring will do very well in the snowy environment, and will tend not to be eaten by the wolves; they will reproduce and produce more offspring carrying the w allele, including white ww individuals. If this happens (and it requires a bit of luck for enough w alleles to survive the first generation and find each other in the snowy environment, so it only happens occasionally), the w allele can make a comeback, the population survives and white bunnies take over the world.

This is of course dependent on the recessive nature of the w gene, which allows it to persist in a hidden fashion in the population even when ww bunnies are being killed off. If you reversed this experiment (start snowy, run for 5 generations, flip to sunny) there are not going to be any B-carrying individuals left in the population (because they would be brown and therefore eaten), and so the bunnies are doomed to die off.

From bunnies to language

Is there a way of interpreting the bunny simulation as being (quite abstractly!) about language? If you wanted to interpret it that way, you would need to decide how to interpret:

- The bunnies (I'd suggest: bunnies correspond to people)
- The trait (for the bunnies it's fur colour; in the language evolution case, could the corresponding trait be presence/absence of language? Or something else?)
- The genes (for the bunnies there is one gene with two variants which affect the trait; is that plausible for your language trait?)
- The wolves (for the bunnies they weed out some traits; what's the language-relevant equivalent?)

The idea here is to think a little bit, hopefully critically, about the idea that language could be a straightforward product of natural selection acting on language genes. These prompts are designed to get people thing about:

- *Granularity. Is the trait "language" vs "no language", with language offering some kind of selective advantage? Or is it more fine-grained, e.g. broken down into the kinds of traits mentioned in the Pinker & Bloom paper discussed in this week's lecture, e.g. does/doesn't distinguish basic ontological categories, does/doesn't allow encoding of distinctions between roles in events, does/doesn't allow talk about intentional states, etc? Or something even more fine-grained?*
- *Selection. Given these levels of granularity, are we happy saying there's a selective advantage for those traits at that granularity? And where does it come from – is it from selective survival of predators and subsequent reproduction, as in the bunny simulation, or something else?*
- *Genetics. Is there any reason to expect a genetic basis for the presence/absence of that trait? Because we're not going to pursue this Pinker & Bloom style explanation*

(the cultural evolutionary account deals with the same phenomenon in a much more straightforward fashion, appealing to directly-observable mechanisms of language change) we're not actually going to spend any time on the course looking at research on the genetics of language, but my impression of this (rapidly-developing) field is that there's no such mapping from genes to specific Pinker & Bloom-type linguistic traits.

People might also notice other mismatches between the bunny simulation and language – e.g. in language selection might be frequency-dependent (it doesn't help to have a new linguistic trait if nobody else shares it), and of course in humans we know that an individual's phenotype depends on their social environment as well as their genes.

Optional additional activities

In the unlikely event you get through all of this, they should have read Pinker (2003) for the Monday lecture, so you could ask them what they thought of that!